RUGVED MHATRE

Austin, TX | rugved.mhatre@nyu.edu | rugvedmhatre.github.io | linkedin.com/in/rugved-mhatre

EDUCATION

New York University, New York 09/2023 – 05/2025
Master of Science in Computer Engineering | ***GPA: 3.8/4.0***
Coursework – High-Performance ML, Deep Learning, Advanced Machine Learning, Advanced Computer Vision, Data Science for Business, Parallel and Customized Computer Architecture (GPU), Data Structures and Algorithms

University of Mumbai, Mumbai 08/2016 – 10/2020
Bachelor of Engineering in Electronics Engineering | ***GPA 3.6/4.0***
Coursework – Database Management Systems, Neural Networks, Digital Image Processing

EXPERIENCE

Annapurna Labs (AWS), Austin | **Software Development Engineer** 10/2025 – Present
- Developed scalable dashboards and automation pipelines to aggregate and analyze hardware telemetry from AWS ML accelerator fleets, improving visibility into system performance and reliability
- Engineered ML-based stress workloads to benchmark next-generation accelerators at Annapurna Labs, driving performance optimization and early detection of architectural bottlenecks

NYU Center for K-12 STEM Education, New York | **Adjunct** 06/2024 – 08/2024, 06/2025 – 08/2025
- Instructed and mentored 150+ high school students in a curriculum focused on Machine Learning and Deep Learning
- Developed hands-on assignments and projects in Numpy, Pandas, Scikit-Learn, TensorFlow, Keras, and PyTorch

New York University, New York | **Graduate Research Assistant** 08/2024 – 05/2025
- Developing simulators for pre-silicon validation and compilers for an accelerator for Privacy-Preserving Machine Learning
- Collaborating with researchers to optimize Full Homomorphic Encryption kernels

Oracle, Mumbai | **Software Development Engineer II (DevOps)** 10/2020 – 06/2023
- Led root cause analysis initiatives across the CI/CD pipeline, identifying and resolving WebLogic deployment issues
- Collaborated with cross-functional teams to streamline workflows and drive deployment efficiency; mentored 5 engineers and led enterprise-wide training on automation tools, enhancing developer productivity
- Awarded Employee of the Month three times for improving CI/CD pipelines, automating 20+ critical DevOps duties, and delivering solutions that boosted developer productivity

Oracle, Mumbai | **Software Development Engineer I (DevOps)** 10/2020 – 09/2022
- Developed scalable RESTful web services using microservices architecture, SOLID principles, and design patterns to enhance core banking operations, leveraging Scrum and Agile methodologies for rapid and efficient delivery
- Implemented a workspace caching system for cloud environments, reducing job setup time from 5 minutes to 30 seconds, boosting resource efficiency and overall operational performance across 100+ Jenkins Jobs
- Designed and integrated a concurrent test execution framework, stress-testing servers with 200+ sessions, and reducing E2E test runtime by 72 hours across 471 test cases for our Java EE Banking Application

SKILLS

Programming Languages: Python, Java, Bash Shell Script, SQL, C, C++, JavaScript
Machine Learning: PyTorch, CUDA, TensorFlow, Keras, Scikit-Learn, LLMs, RAG, LangGraph, Agentic AI, GenAI

PROJECTS

Multimodal Sentiment Analysis using Transformers
- Researched multimodal fusion techniques to mix audio, text, and video data for improved sentiment analysis performance
- Achieved 75% accuracy on CMU-MOSI and CMU-MOSEI datasets using Early Fusion Transformers, Multimodal Transformer models, and experimented with innovative approaches such as the Tensor Fusion Model

Efficient Vision Transformers: Once-Train Optimization
- Developed a once-trained network framework for Vision Transformers (ViTs) to enable efficient deployment across diverse hardware platforms, utilizing FlashAttention, Neural Architecture Search, Pruning, Distillation, and Quantization
- Conducted in-depth profiling of GPU-accelerated applications to identify performance bottlenecks, memory usage patterns, and computational inefficiencies, along with benchmarking methodologies to measure inference performance
- Achieved a model with 95% fewer parameters and 75% lower memory used with minimum performance tradeoffs

Continual Learning for Autonomous Vehicles
- Developed a CNN continual learning system for autonomous vehicle steering control using lifelong learning techniques, resulting in the model adapting to new data without forgetting previous knowledge
- Formulated a novel method of Temporal Consistency Regularization with a balanced buffer to enhance prediction accuracy
- along with smoothness of drive, reducing MSE by 18% while also reducing memory used by 25%